

NIRIKSHAK AI — Statutory Compliance Audit Specification

Official Platform Rules, Mathematical Evaluation Logic & Dataset Metrics (75,501 Works)

1. Executive Overview

NIRIKSHAK AI provides automated statutory compliance monitoring for the Member of Parliament Local Area Development Scheme (MPLADS). The platform audits a real-world dataset of **75,501 works** across India to ensure strict adherence to administrative procedures, financial ceilings, and execution chronologies.

2. Status Categorization & Dataset Breakdown

Status Verdict	Definition & Logic	Works Count	Percentage
Compliant (■)	0 rule violations AND status is COMPLETED with physical certificate.	57,282	75.9%
Under Review (■)	0 rule violations AND work is active (SANCTIONED, IN_PROGRESS).	15,477	20.5%
Non-Compliant (■)	Triggers 1 or more statutory rule violations.	2,742	3.6%

3. Statutory Audit Rules & Formulas

Rule 1: Expenditure Before Sanction (EXP_BEFORE_SANCTION) — CRITICAL ■

Flags disbursements prior to administrative sanction date/approval or when status is RECOMMENDED_ONLY.

Formula: $\text{exp_amount} > 0 \text{ AND } (\text{sanctioned_amount} == 0 \text{ OR } \text{status} == \text{'RECOMMENDED_ONLY'} \text{ OR } \text{exp_date} < \text{sanction_date})$

Dataset Result: 2,077 works flagged (97.2% compliance rate).

Rule 2: Expenditure Exceeding Sanction (EXP_EXCEEDS_SANCTION) — HIGH ■

Flags cases where actual expenditure exceeds approved sanction limit.

Formula: $(\text{exp_amount} > \text{sanctioned_amount} \text{ AND } \text{sanctioned_amount} > 0) \text{ OR } \text{cost_overrun_pct} > 5.0\%$

Dataset Result: 621 works flagged (99.2% compliance rate).

Rule 3: Excessive Execution Delay (EXCESSIVE_DELAY) — HIGH ■

Flags projects pending >365 days post-sanction without completion.

Formula: $(\text{days_since_sanction} > 365 \text{ OR } \text{is_delayed} == 1) \text{ AND } \text{status} != \text{'COMPLETED'}$

Rule 4: Missing Physical Completion Certificate (MISSING_COMPLETION_CERT) — HIGH ■

Flags works with 100% fund disbursement but missing completion certificate.

Formula: $\text{exp_amount} \geq \text{sanctioned_amount} \text{ AND } \text{sanctioned_amount} > 0 \text{ AND } \text{status} != \text{'COMPLETED'}$

Rule 5: Financial vs Physical Mismatch (FINANCIAL_PHYSICAL_MISMATCH) — MEDIUM ■

Flags high fund utilization (>85%) with incomplete physical progress.

Formula: $(\text{financial_execution_rate} \geq 85.0\% \text{ OR } \text{exp_amount} / \text{sanctioned_amount} \geq 0.85) \text{ AND } \text{status} != \text{'COMPLETED'}$

Dataset Result: 29 works flagged.

Rule 6: Duplicate Disbursement Pattern (DUPLICATE_PAYMENT_PATTERN) — HIGH ■

Flags duplicate payment claims across matching constituency records.

Formula: Duplicate Match on (constituency, sanctioned_amount, work_category)

Dataset Result: 15 works flagged.

4. Financial Year Compliance Breakdown

Financial Year	Total Works	Compliant	Under Review	Non-Compliant	Health Score
All Financial Years	75,501	57,282	15,477	2,742	90.4
FY 2025-26	33,398	25,410	6,712	1,276	91.2
FY 2024-25	10,463	7,921	2,110	432	89.8
FY 2023-24	1,132	861	228	43	91.5
FY 2026-27	11,699	8,901	2,315	483	90.1